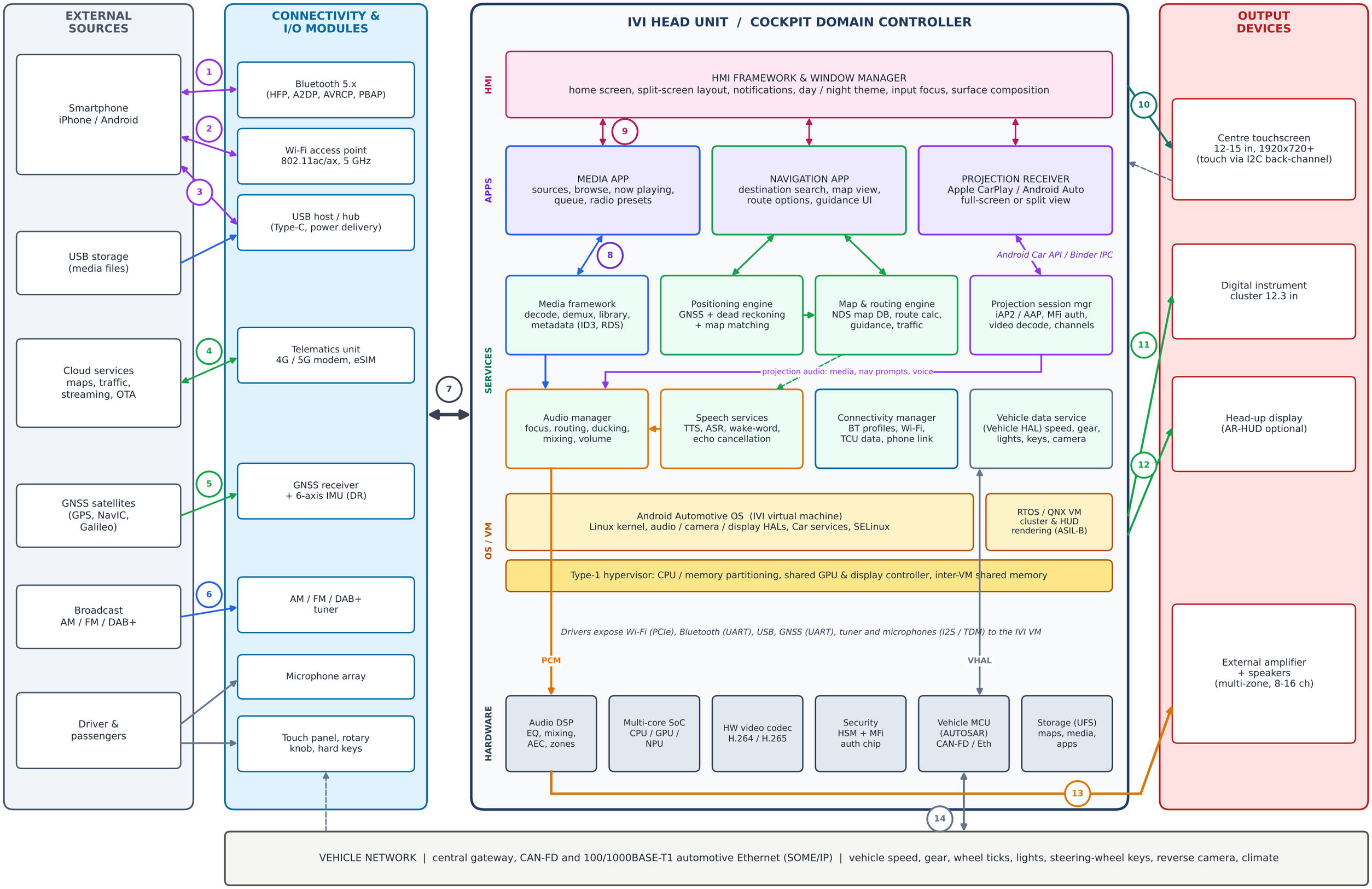
Conceptual Architecture of an In-Vehicle Infotainment (IVI) System

Integrated media, navigation and smartphone projection (Apple CarPlay / Android Auto) on a cockpit domain controller



DATA-FLOW KEY

- Media / audio source data
- Navigation data
- Projection session
- Audio output
- Video / display
- Control & vehicle data

INTERFACE ANNOTATIONS

1. Bluetooth: pairing, iAP2 / RFCOMM session start, HFP calls, A2DP audio, AVRCP control, PBAP contacts
2. Wi-Fi 5 GHz (802.11ac): wireless CarPlay / Android Auto video, audio and control over TCP/IP
3. USB 2.0 HS: iAP2 + CDC-NCM (wired CarPlay), AOA (wired Android Auto), MSC (USB media), charging
4. LTE / 5G: online map tiles and updates (NDS.Live), live traffic (TPEG), streaming, OTA software
5. GNSS L1 / L5 signals: raw position and time; IMU and wheel ticks add dead reckoning in tunnels
6. RF broadcast: AM / FM with RDS, DAB+ with DLS metadata and slideshow images
7. Internal I/O: PCIe (Wi-Fi), UART / USB (BT), USB, UART (GNSS), I2S / TDM (tuner audio, microphones)
8. Android Car API / Binder IPC between apps and services: commands down, state and data up
9. Surface composition and input events between apps and the HMI / window manager
10. GMSL2 / FPD-Link serial video to centre display, touch data returned over the link's I2C back-channel
11. Inter-VM shared memory: turn-by-turn, media and phone info rendered by RTOS on the cluster
12. Maneuver arrow, distance, lane guidance and speed limit to HUD over serial video link
13. A2B (Automotive Audio Bus): multichannel PCM from audio DSP to amplifier, volume and zone control
14. CAN-FD / SOME/IP via vehicle MCU into Vehicle HAL: speed, gear, lights, steering-wheel keys, camera

Dashed arrows: control / feedback